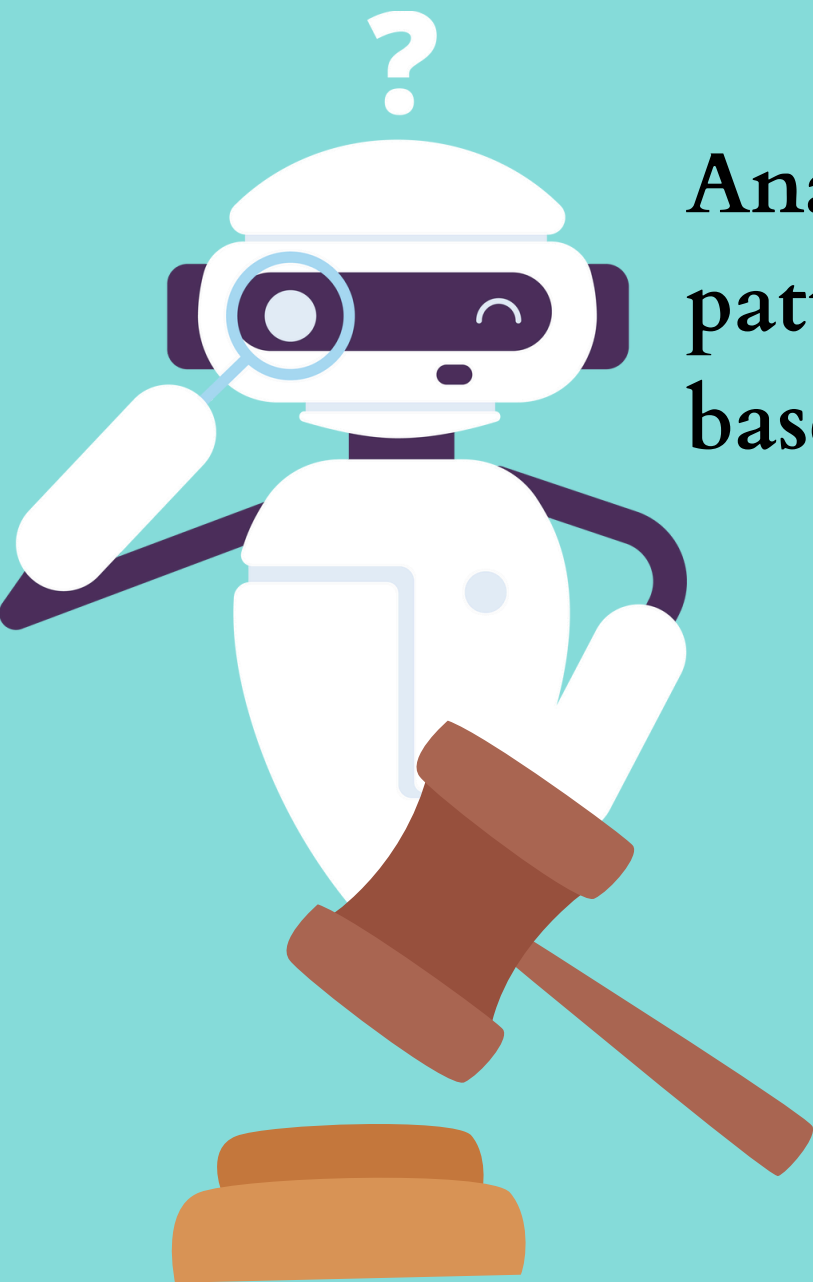
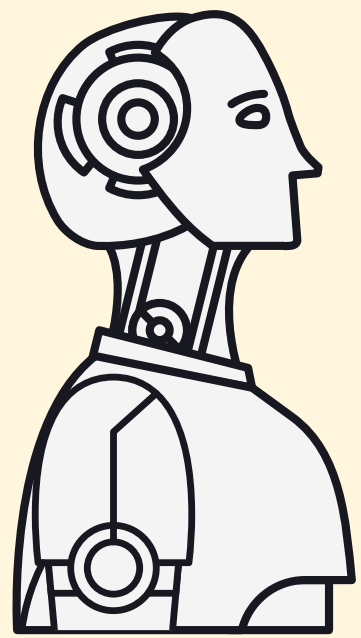




# Analyzing the criminal cases to find pattern in outcome of bail application based on IPCs

By Kushagra and Sudipta  
Under supervision of Prof. S M Bendre



# Introduction

- The Indian judicial system is one of the oldest and world's largest judiciary systems. All these courts give judgment, mostly following the Indian Penal Code and It is their duty to protect and uphold the rights under the constitution.
- Despite being the most dependable and powerful institution for getting justice for billions of Indians, It is very slow in issuing verdicts. Millions of cases are trapped in legal logjams (Mahadik et. al. 2018).
- Many studies have shown the success of Machine Learning Algorithms in prediction of the probability of outcome of appeal decisions. Few of which are, prediction of judicial decisions on Germany's tax law cases, predicting jurisdiction of the European Court of Human Rights, predicting the outcomes in the higher courts of Turkey, Prediction of Legal Decisions of the Indian Central Information Commission (Waltl et. al. 2017, Aletras et. al. 2016, Mumcuoglu et. al. 2021, Kalia et. al. 2022) .

# OVERVIEW

- The judgements on applications of bail (of criminal cases), also called an order of exhibit were available to us as dataset. We collected these from [ecourt.gov.in](http://ecourt.gov.in).
- In this study, we tried to find a pattern in Order of Exhibit, whether bail granted or, rejected for the criminal cases.
- To explore and analyze the dataset, text and IPC sections are being extracted out of it.
- And we have performed Exploratory Data Analysis and modeling, like Naive Bayes, k-Nearest Neighbor, Neural Network, Bagging on the dataset.
- We have used Python for the whole exploration and modeling, mostly nltk, re, pandas, scikit learn, keras, Tensorflow libraries were used.

# **Material & Methods**

# EXTRACTION & PROCESSING OF FEATURES

1      Cri. B.A. No. 2941/17.

## Order below Exh. 1 in Cri. Bail Application No. 2941/2017 :-

This is an application filed by the applicants under Section 438 of the Code of Criminal Procedure, praying for releasing them on anticipatory bail in connection with C.R. No.106/17 registered with Ranjangaon police station, Dist. Pune, under Sections 498-A, 494, 406, 313, 323, 504 and 506 read with section 34 of the Indian Penal Code.

2] The applicants have submitted that as per prosecution story, the complainant Sou. Hira alias Priyanka lodged complaint alleging that the applicants pressurized her and got her child aborted. Persons from her matrimonial house were illtreating her and they were asking to bring money from her parents. It is further alleged by the complainant that on 19-12-13, she was pregnant for second time and again she was asked for abortion.

Fig 1 A part of resolved Bail Application

|     | 405 | 403 | 414 | 457 | 500 | 312 | 412 | 328 | 418 | 504 | ... | 212 | 318 | 407 | 177 | 203 | 279 | 286 | OTHER | 204 | RESULT |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|-------|-----|--------|
| 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 0      |
| 1   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 0      |
| 2   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 1   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 0      |
| 3   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 0      |
| 4   | 0   | 0   | 0   | 1   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 0      |
| ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ... | ...   | ... | ...    |
| 509 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 1      |
| 510 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 1   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 1      |
| 511 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 1   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 1      |
| 512 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 1   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 1      |
| 513 | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 1   | ... | 0   | 0   | 0   | 0   | 0   | 0   | 0   | 0     | 0   | 1      |

Fig: Dataset in Input format for modeling

|     | TEXT                                              | CATEGORY |
|-----|---------------------------------------------------|----------|
| 0   | bail appln court additional session judge pune... | 0.0      |
| 1   | bail appln court additional session judge pune... | 0.0      |
| 2   | bail appln chandani gore state kondhwa police ... | 0.0      |
| 3   | order bail atish balram raste state applicant ... | 0.0      |
| 4   | mhpu criminal misc order perused heard learned... | 0.0      |
| ... | ...                                               | ...      |
| 509 | order bail filed applicant section code crimin... | 1.0      |
| 510 | order criminal bail filed accused section code... | 1.0      |
| 511 | order bail filed applicant section code crimin... | 1.0      |
| 512 | order bail filed applicant harshal sangeet kas... | 1.0      |
| 513 | order bail filed applicant section code crimin... | 1.0      |

Table: Data frame with string of Text

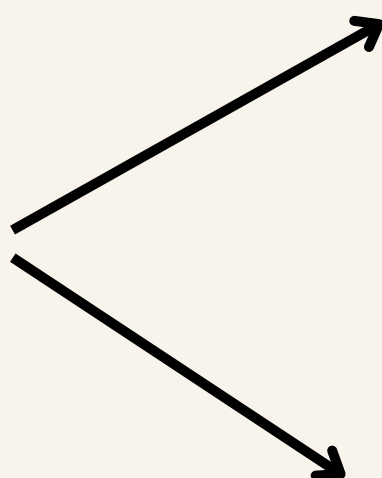
# Data Preparation (Text)

1. UNNECESSARY SYMBOLS ARE REMOVED
2. STOP WORDS ARE REMOVED
3. REMAINING WORDS IN TEXT ARE LEMMATIZED

METHODS USED FOR  
FEATURE  
EXTRACTION FROM  
CLEANED TEXT

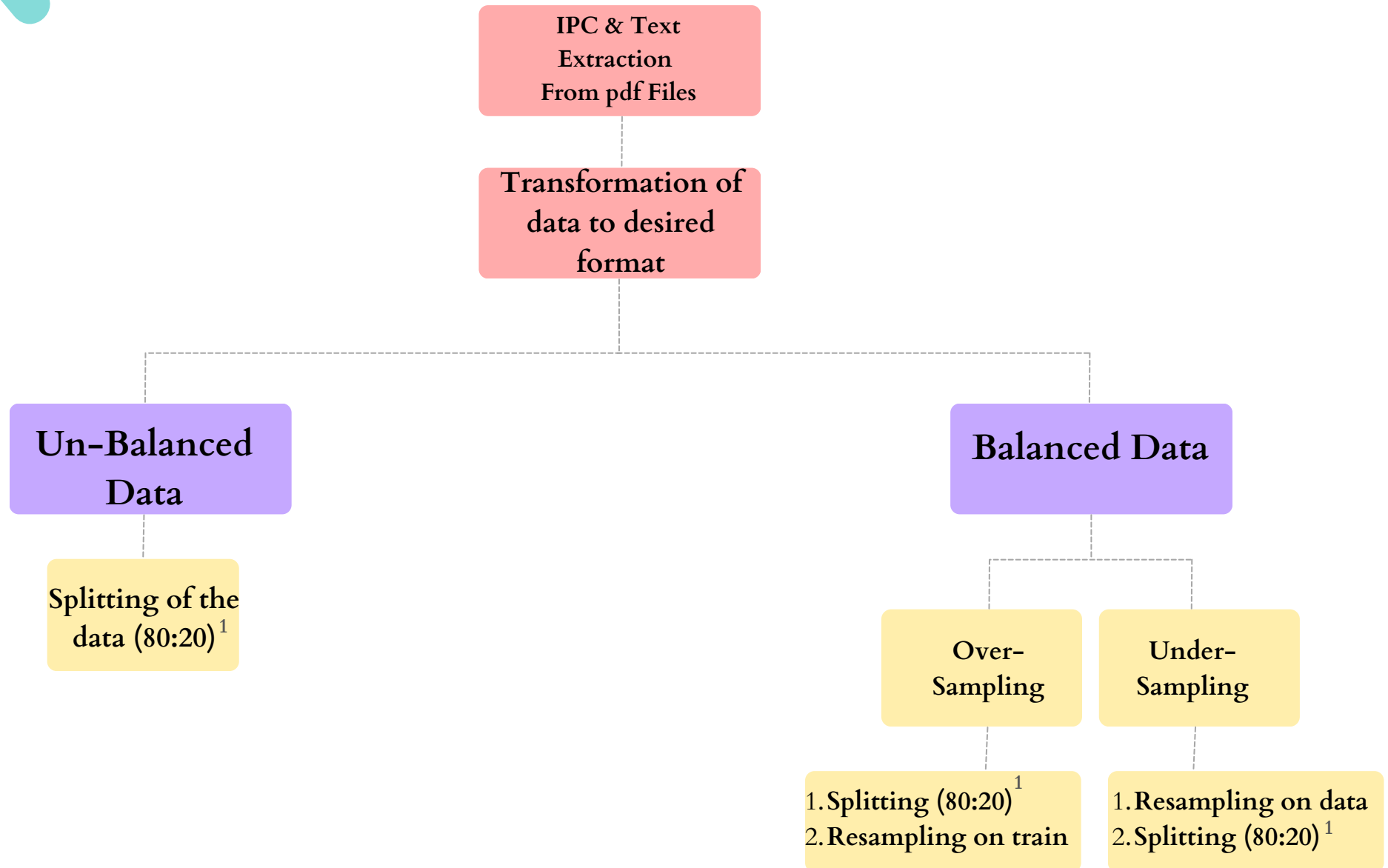
COUNT VECTORIZER

TF-IDF( (TERM FREQUENCY INVERSE  
DOCUMENT FREQUENCY)





# Data Preparation (IPCs)



<sup>1</sup> 80:20 = { 80% train dataset  
20% test dataset



# IPC Data Set Distribution

**DATASET**

**ORIGINAL  
(UNBALANCED)**

| Dataset       | Bail | No Bail |
|---------------|------|---------|
| Total Dataset | 344  | 170     |
| Training      | 271  | 140     |
| Testing       | 73   | 30      |

**OVER SAMPLING  
(BALANCED)**

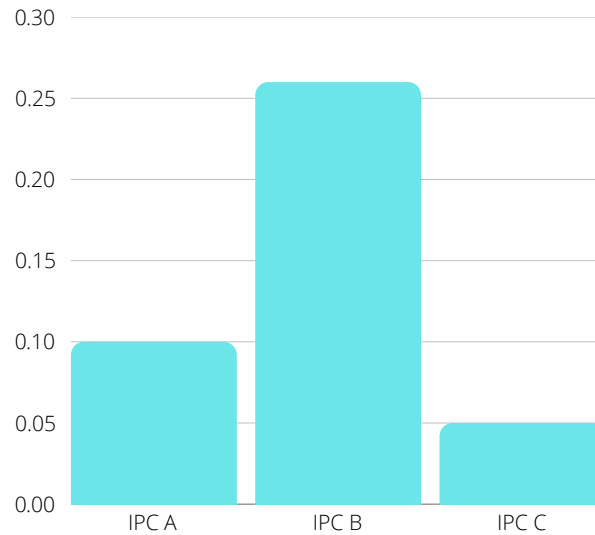
| Dataset       | Bail | No Bail |
|---------------|------|---------|
| Total Dataset | 344  | 301     |
| Training      | 271  | 271     |
| Testing       | 73   | 30      |

**UNDER SAMPLING  
(BALANCED)**

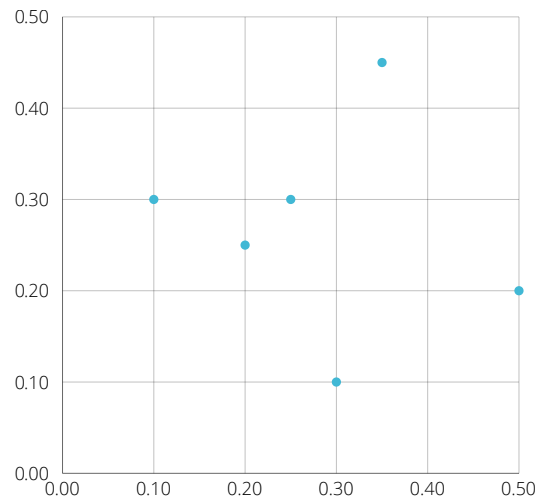
| Dataset       | Bail | No Bail |
|---------------|------|---------|
| Total Dataset | 170  | 170     |
| Training      | 140  | 140     |
| Testing       | 30   | 30      |

# Exploratory Data Analysis:

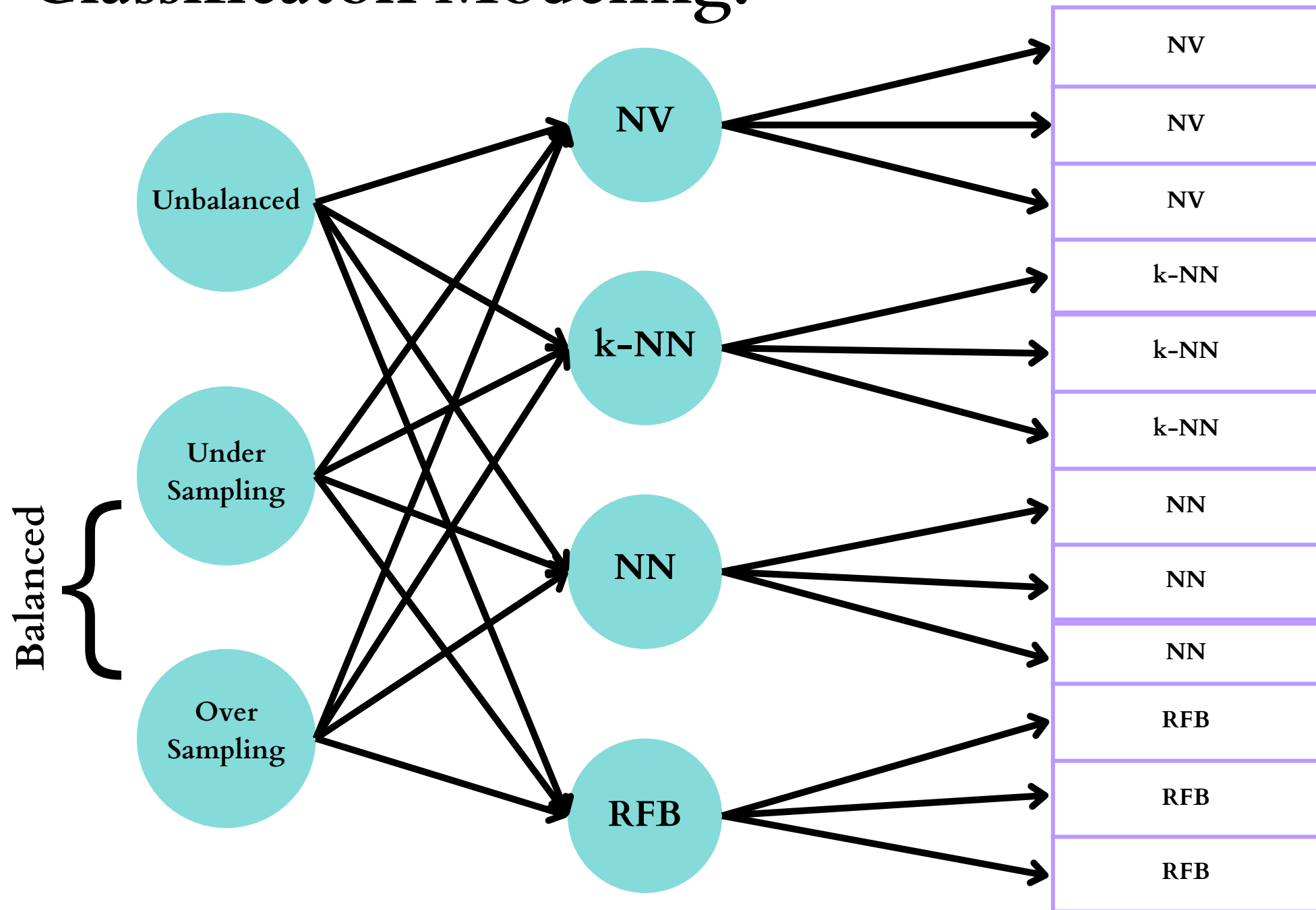
- Bar plots of IPCs in all Bail & all No Bail files,



- Scatter plot of IPCs in all Bail vs all No Bail files,



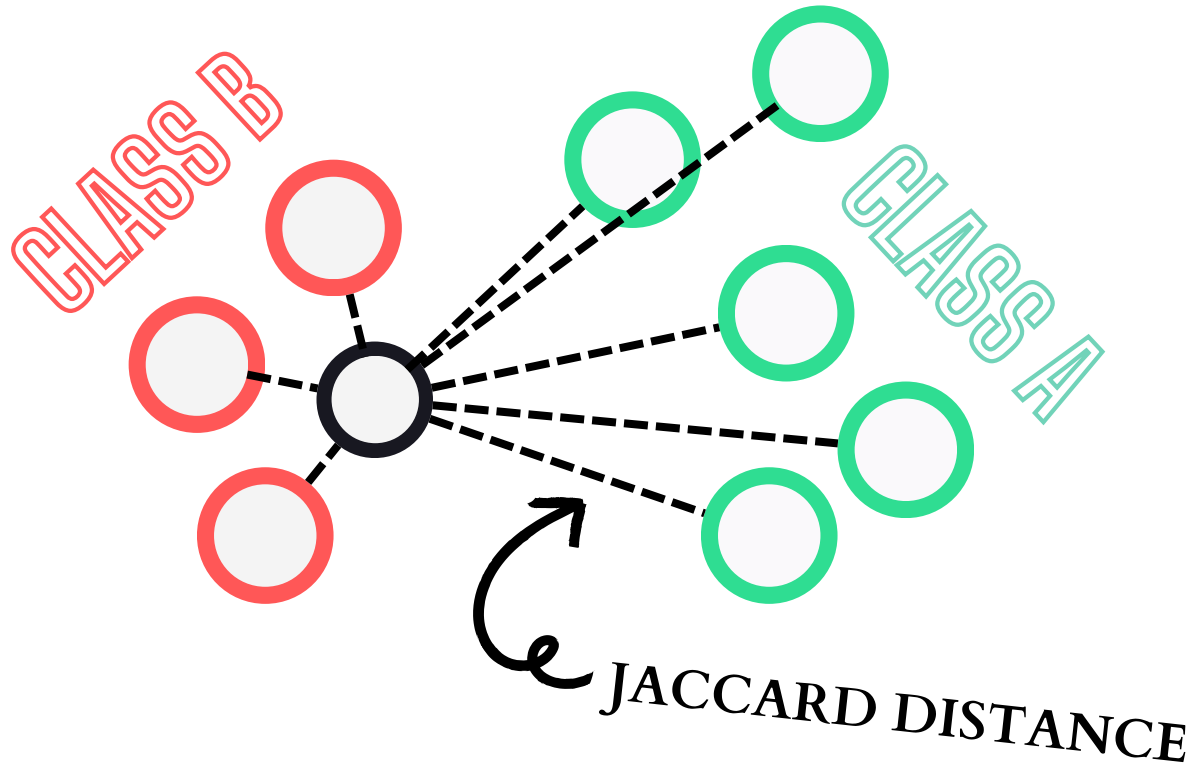
# Classification Modeling:



# NAIVE BAYES CLASSIFIER

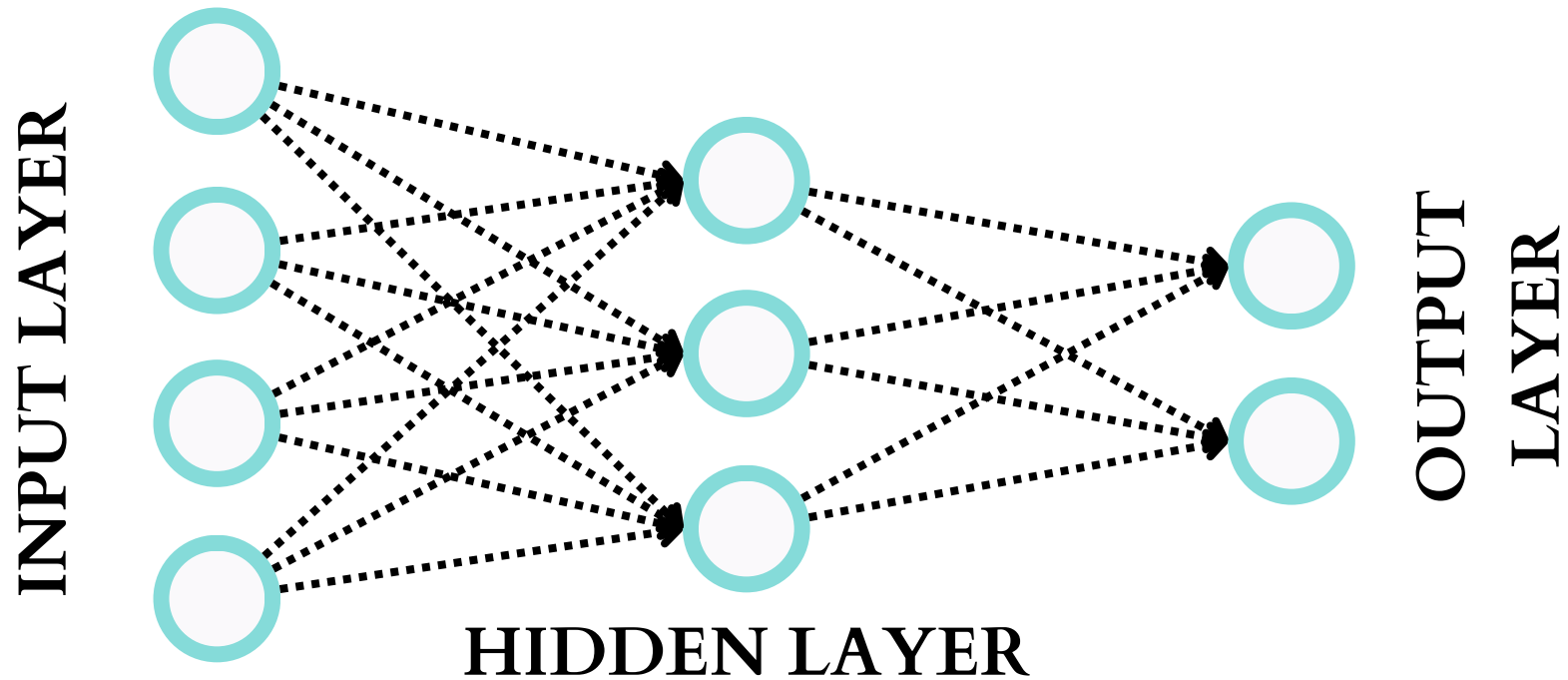
$$P(\text{class} | \text{data}) = \frac{P(\text{data} | \text{class}) \times P(\text{class})}{P(\text{data})}$$

# K- NEAREST NEIGHBOUR



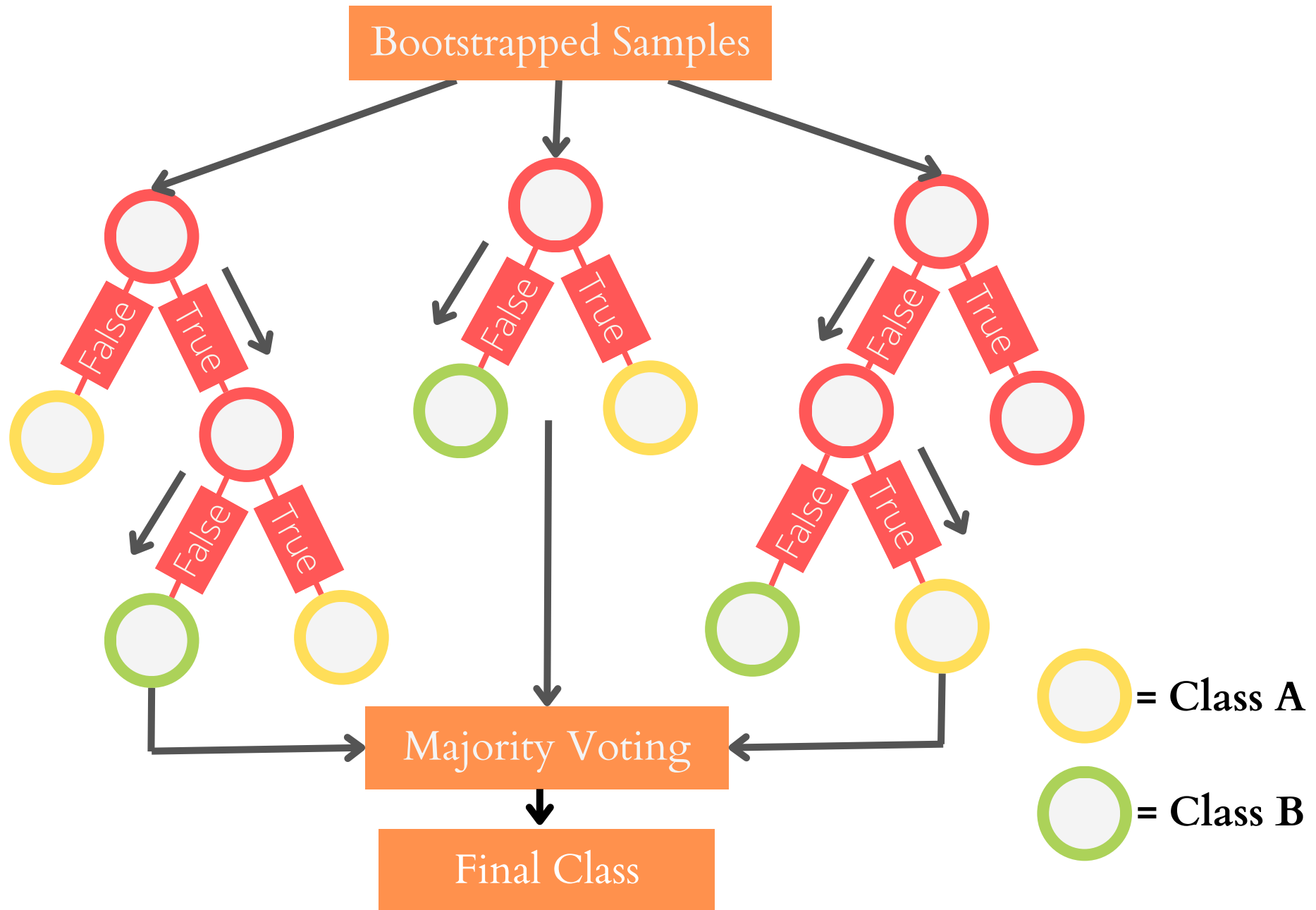
- We have plotted k-value vs error rate, to get 'best' k-value,
- We have used Jaccard Distance for calculating the distance matrix.

# NEURAL NETWORK



- We have optimized the NN architecture with Keras and Tensorflow.

# RANDOM FOREST (BAGGING)



# EVALUATION METRICS

In this study, all the confusion matrix are in the below illustrated format,

|                    |                        | <b>P R E D I C T E D</b> |                     |
|--------------------|------------------------|--------------------------|---------------------|
|                    |                        | No Bail(0)<br>Negative   | Bail(1)<br>Positive |
| <b>A C T U A L</b> | No Bail(0)<br>Negative | True Negative            | False Positive      |
|                    | Bail(1)<br>Positive    | False Negative           | True Positive       |

Fig: Confusion Matrix

(tn = true negative; tp = true positive; fp = false positive; fn = false negative)



# CONFUSION MATRIX METRICS

## 1. Sensitivity/Recall:

$$\text{recall} = \frac{tp}{tp + fn}$$

## 2. Specificity/TNR

$$TNR = \frac{tn}{tn + fp}$$

## 3. Accuracy

$$\text{accuracy}(y, \hat{y}) = \frac{1}{n_{\text{samples}}} \sum_{i=0}^{n_{\text{samples}}-1} 1(\hat{y}_i = y_i)$$

where  $y$  and  $\hat{y}$  are actual and predicted classes for the  $i$ th case.

## 5. F1-Score

$$F_{\beta} = (1 + \beta^2) \frac{\text{precision} \times \text{recall}}{\beta^2 \text{precision} + \text{recall}}$$

Here  $\beta$  is 1 for f1 score

## 4. Precision

$$\text{precision} = \frac{tp}{tp + fp}$$

## 6. Error Rate

$$\text{Error Rate} = \frac{\sum_{i=1}^n I(a_i \neq b_i)}{n}$$

Where,  $a$  = actual class in test data set,  
 $b$  = predicted class,  
 $I$  = Indicator Function.

# **Result & Discussion**

# MODELING WITH TEXT DATA

*WHAT IS THIS TEXT ABOUT EXACTLY?*

*IT IS A SUMMARY OF THE JUDGEMENT GIVEN.*

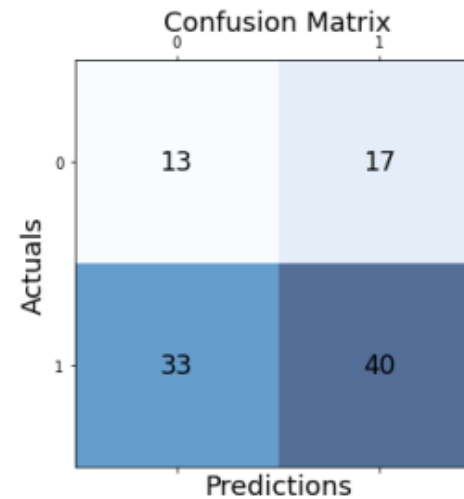
*WHAT MODELS ARE WE  
USING?*

1. *NAIVE BAYES*
2. *k-NN*

# RESULTS FROM TEXT DATA

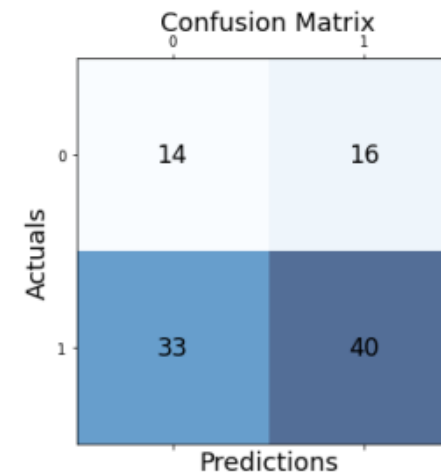
## 1. NAIVE BAYES

### 1.1 USING COUNT VECTORIZER



Sensitivity : 0.5479  
Specificity : 0.4333  
Accuracy : 0.5145  
F<sub>1</sub>-Score : 0.6153

### 1.2 USING TF-IDF VECTORIZER

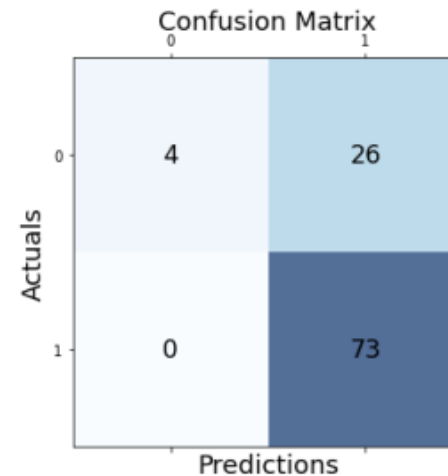


Sensitivity : 0.5479  
Specificity : 0.4666  
Accuracy : 0.5242  
F<sub>1</sub>-Score : 0.6201

# RESULTS FROM TEXT DATA

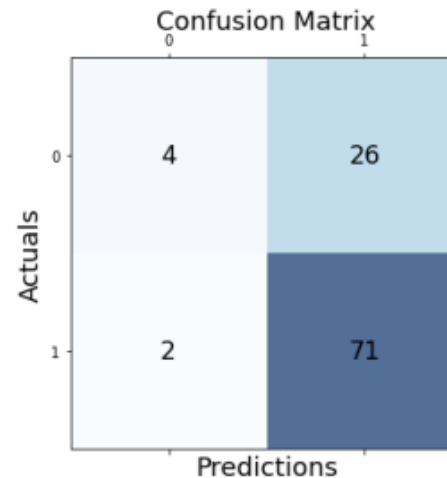
## 2. k - Nearest Neighbour

### 2.1 USING COUNT VECTORIZER



Sensitivity : 1  
Specificity : 0.1333  
Accuracy : 0.7475  
F<sub>1</sub>-Score : 0.8487

### 2.2 USING TF-IDF VECTORIZER



Sensitivity : 0.9726  
Specificity : 0.1333  
Accuracy : 0.7281  
F<sub>1</sub>-Score : 0.833

# CONCERNS IN USING TEXT

1. **Partial or No summary.**
2. **Spelling Mistakes and abbreviations.**
3. **Few words from other Languages.**
4. **Of no practical use.**
5. **Low accuracy and specificity of models.**



# EXPLORATORY DATA ANALYSIS

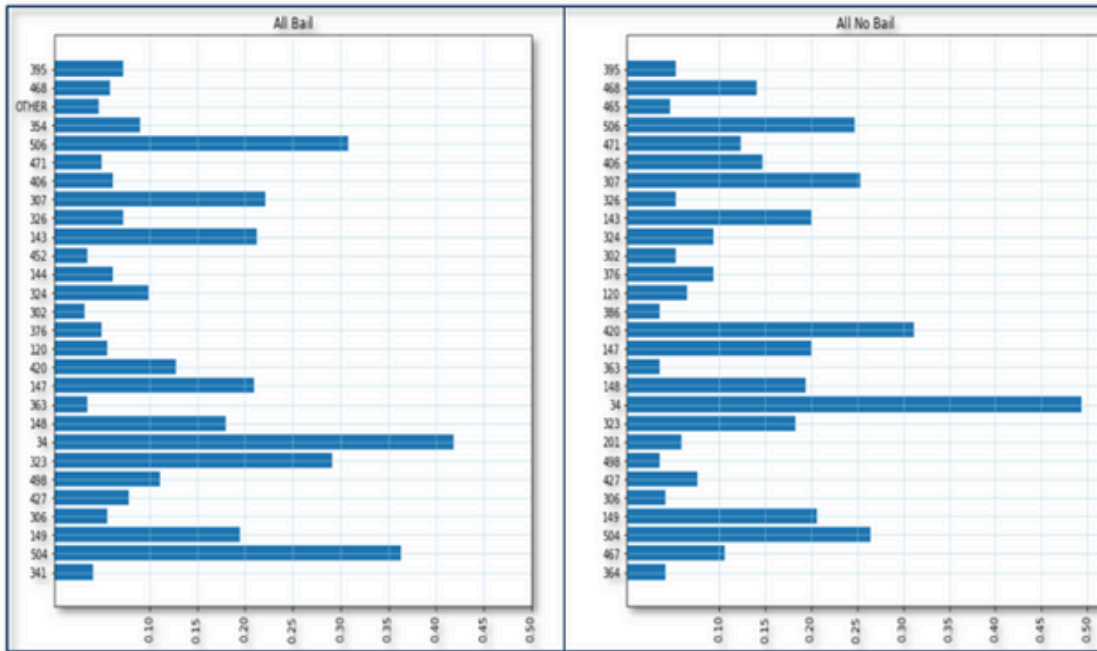


Fig 7: IPCs in all Bail files

Fig 8: IPCs in all No Bail files

i. Bar plots of IPCs in all Bail & all No Bail files

ii. Scatter plot of IPCs in all Bail vs all No Bail files

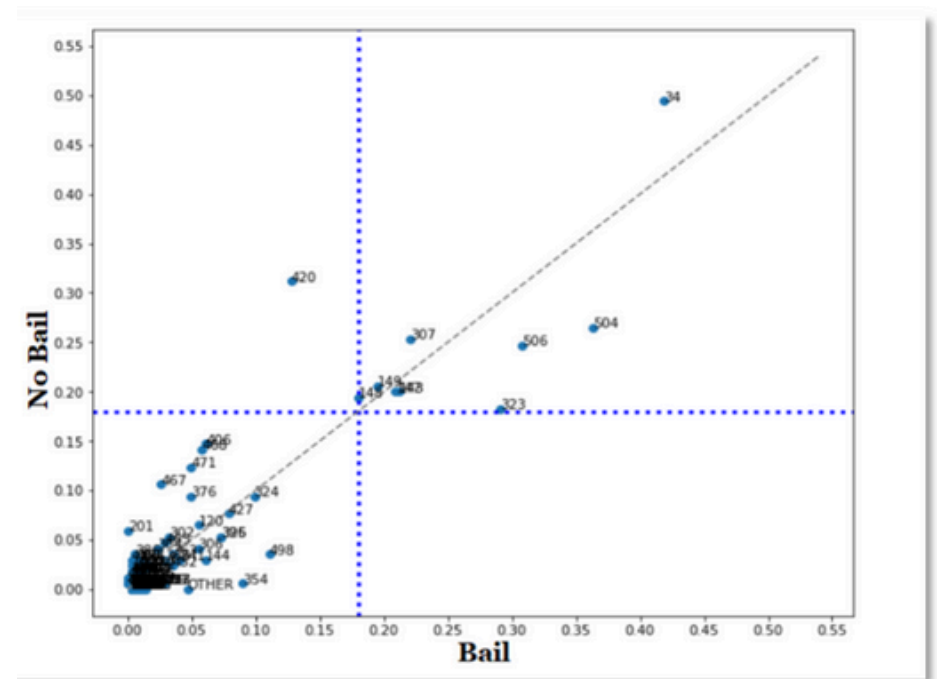


Fig 9: Scatter Plot Bail vs. No Bail

# SUMMARIZING EDA RESULTS

| Cases                          | IPCs                                                            |
|--------------------------------|-----------------------------------------------------------------|
| Present in both Bail & No Bail | 427, 395, 363, 326, 323, 324, 307, 302, 149, 148, 147, 143, 120 |
| Only in Bail                   | 452, 354, 341, 144                                              |
| Only in No Bail                | 467, 465, 386, 364, 302, 201                                    |
| More in Bail & less in No Bail | 504, 498, 323 ,506                                              |
| More in No Bail & less in Bail | 468, 471, 420, 406, 376                                         |

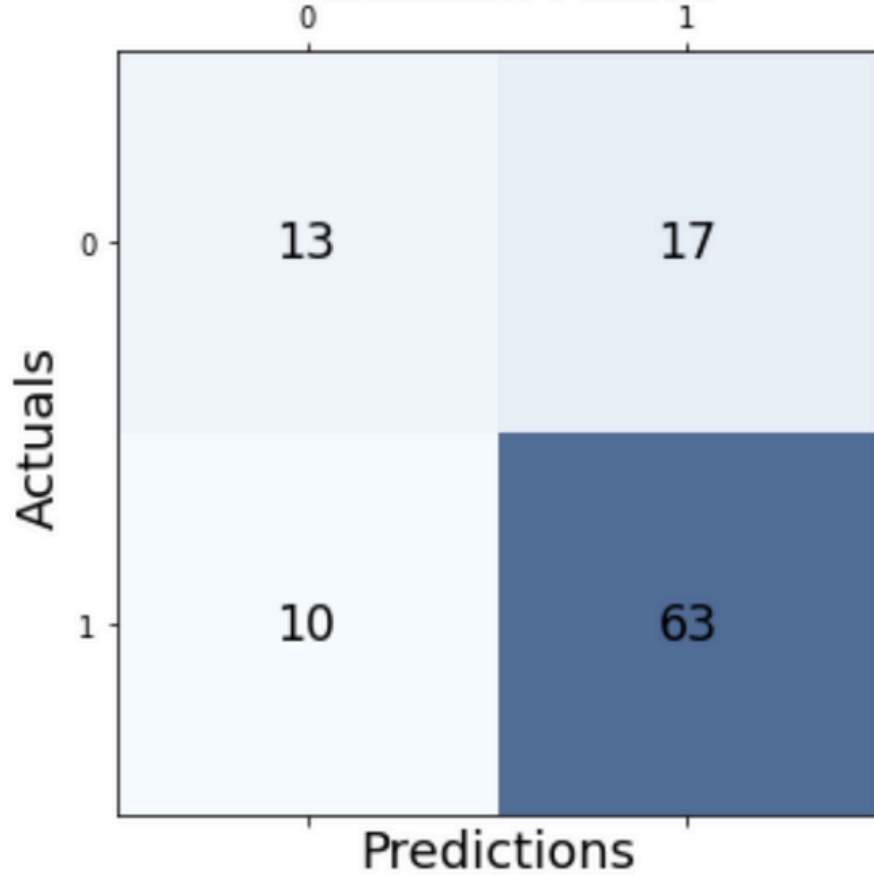
Table: Summary of IPCs





# MODELS USING ORIGINAL DATA

Confusion Matrix



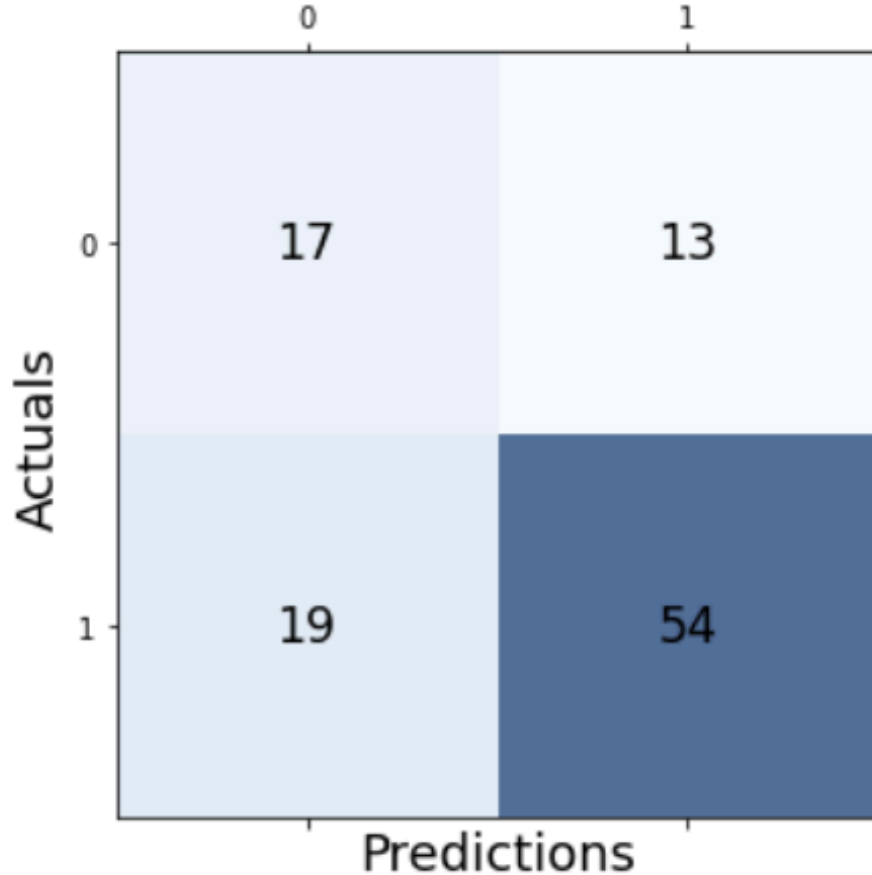
k-NN (Unbalanced)



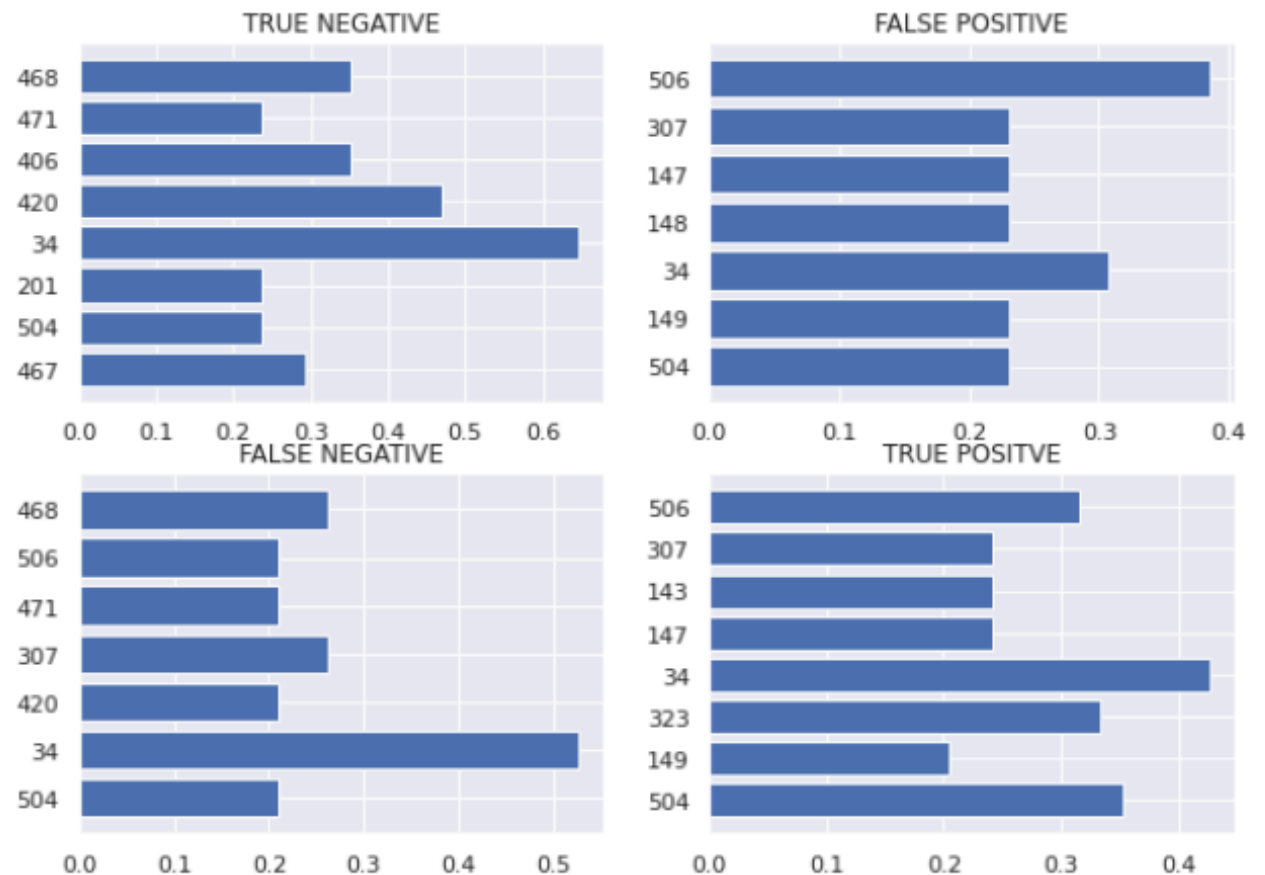
# RANDOM FOREST(BAGGING)

## USING ORIGINAL DATA

Confusion Matrix

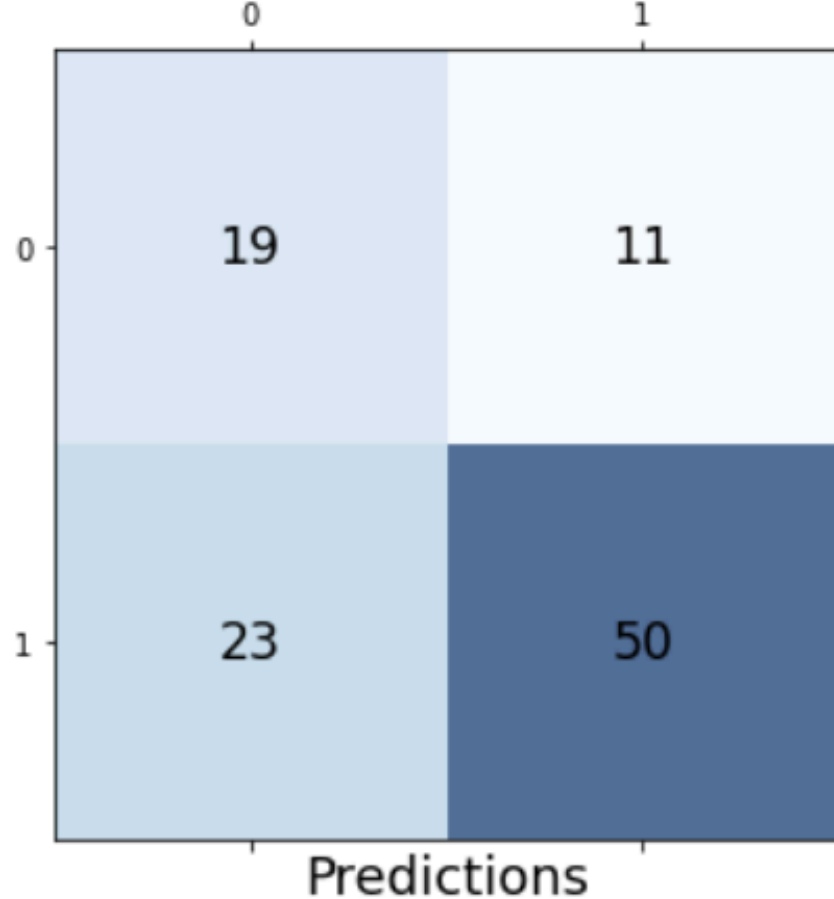


Random Forest (Unbalanced dataset)



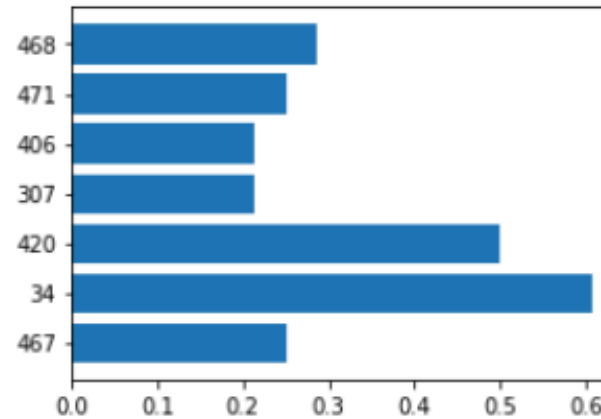
# MODELS USING OVERSAMPLED DATA

Confusion Matrix

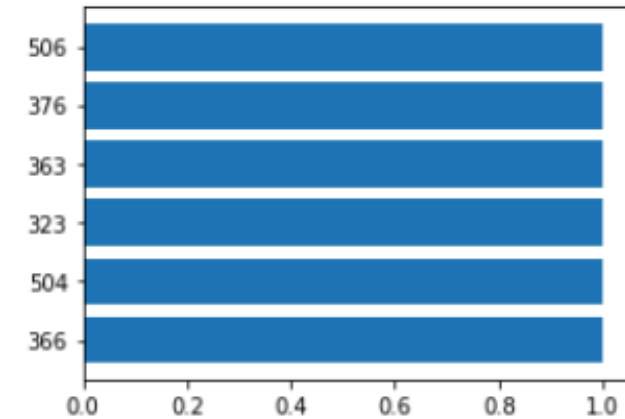


k-NN (Over Sampling)

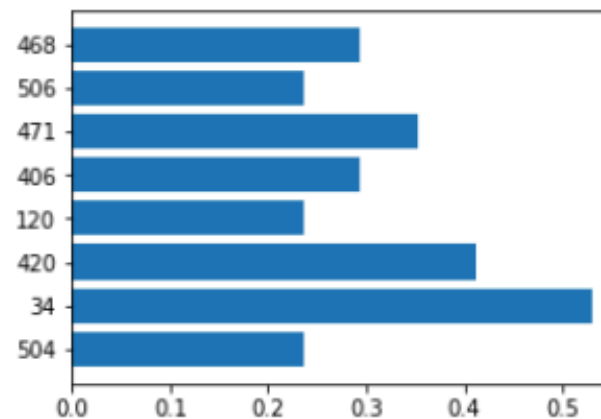
TRUE NEGATIVE



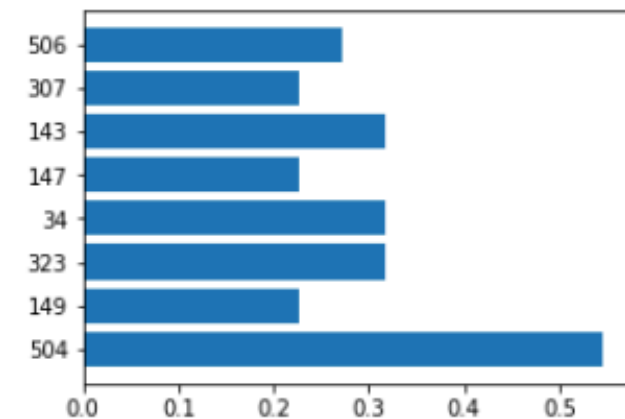
FALSE POSITIVE



FALSE NEGATIVE

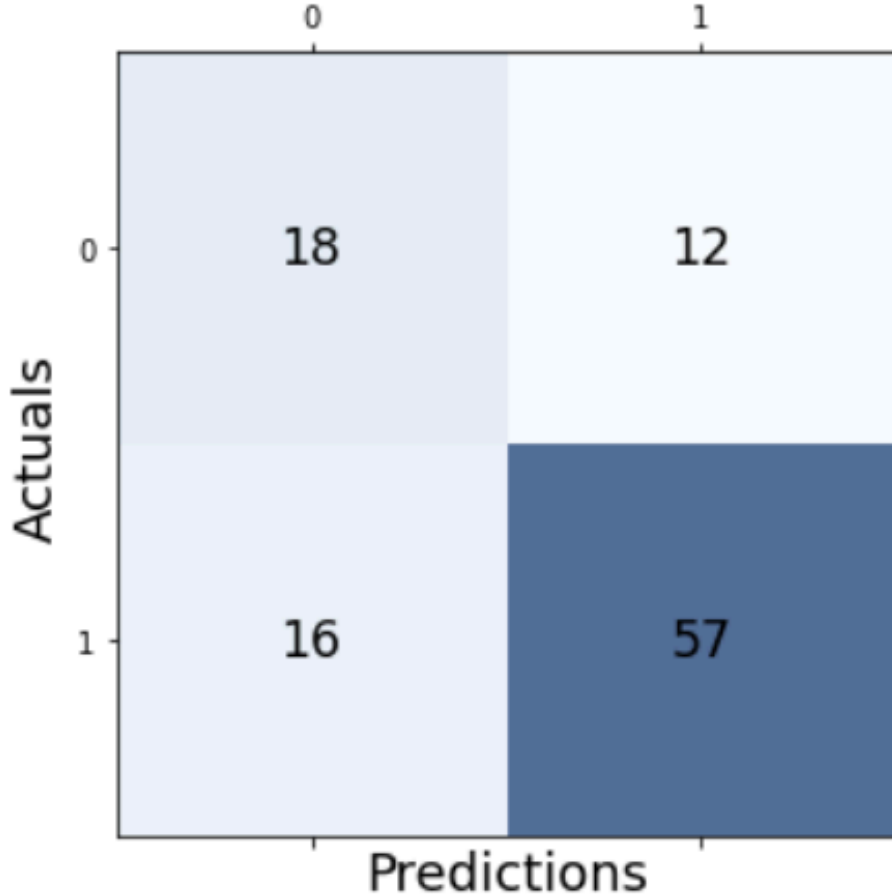


TRUE POSITIVE

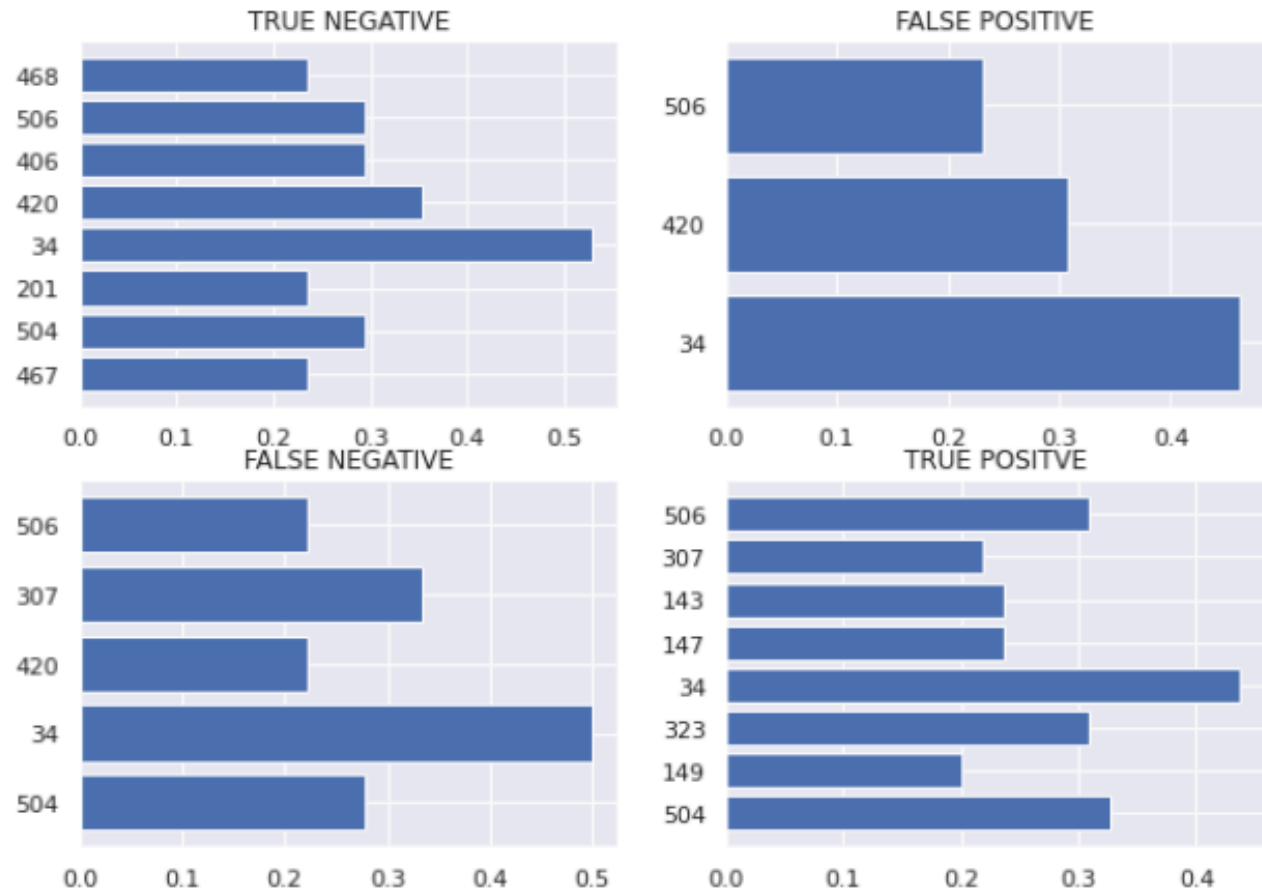


# MODELS USING OVERSAMPLED DATA

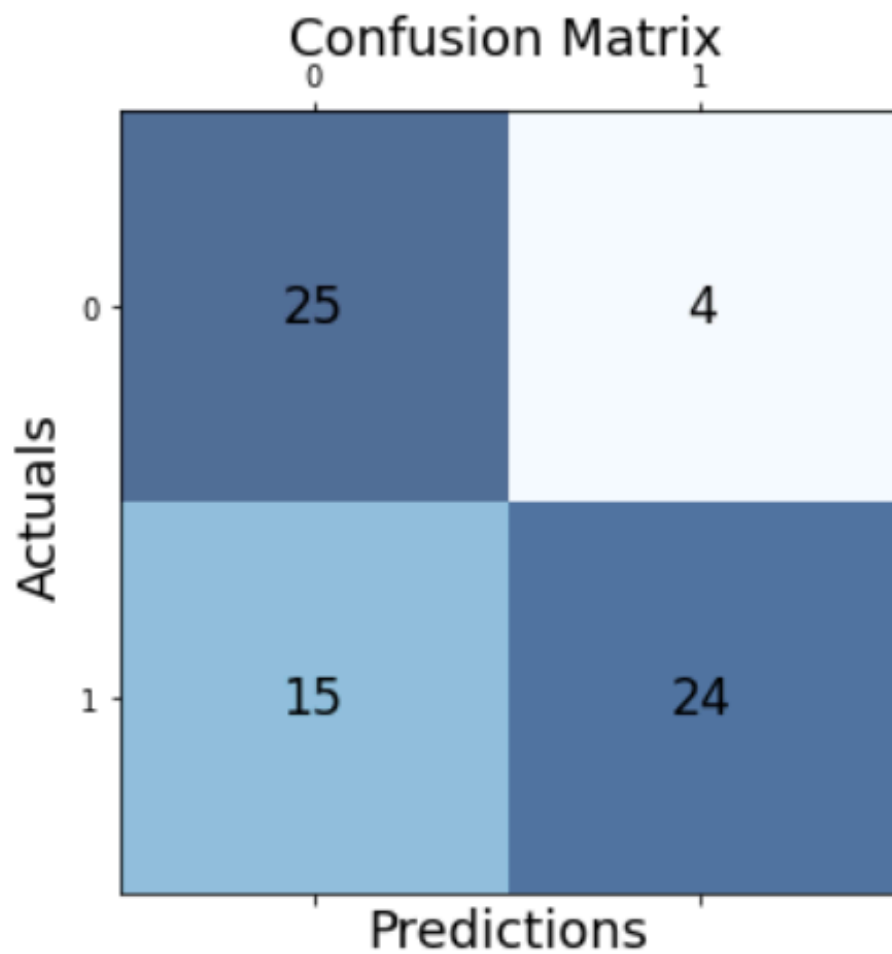
Confusion Matrix



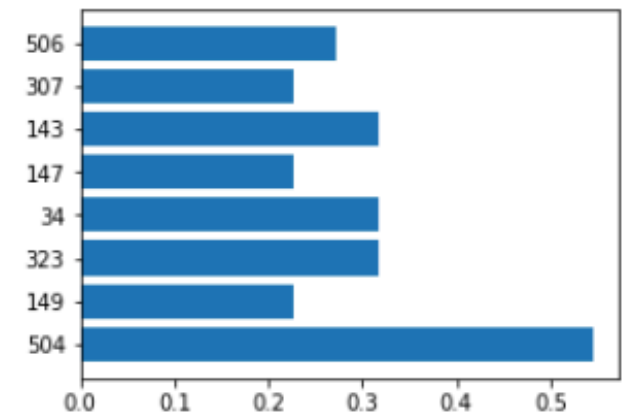
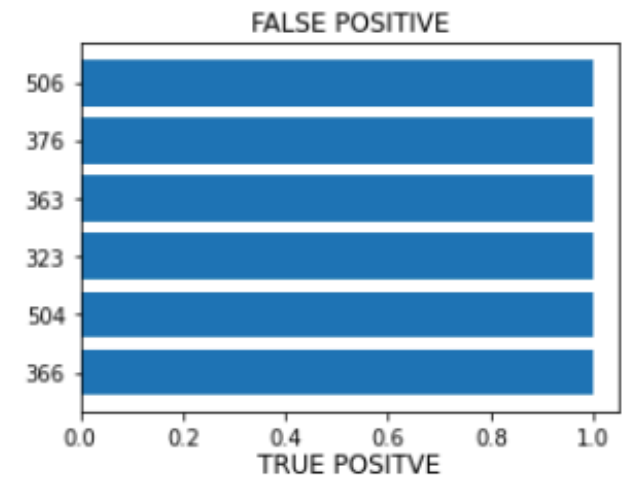
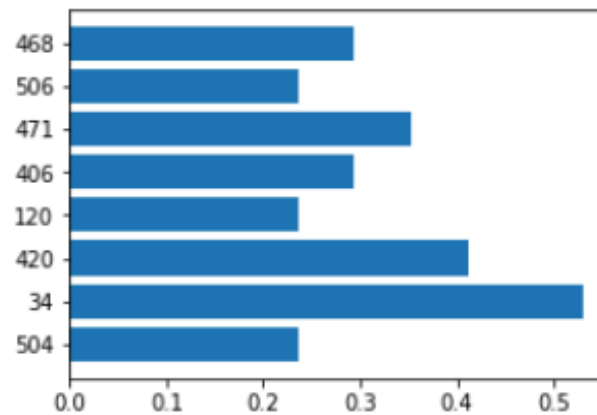
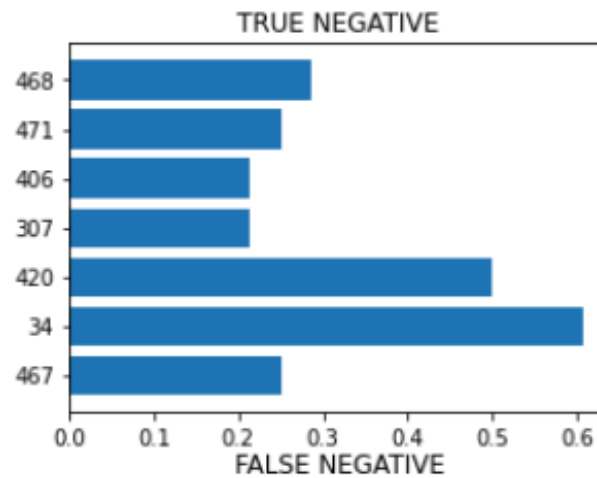
NN (Oversampling)



# MODELS USING UNDERSAMPLED DATA

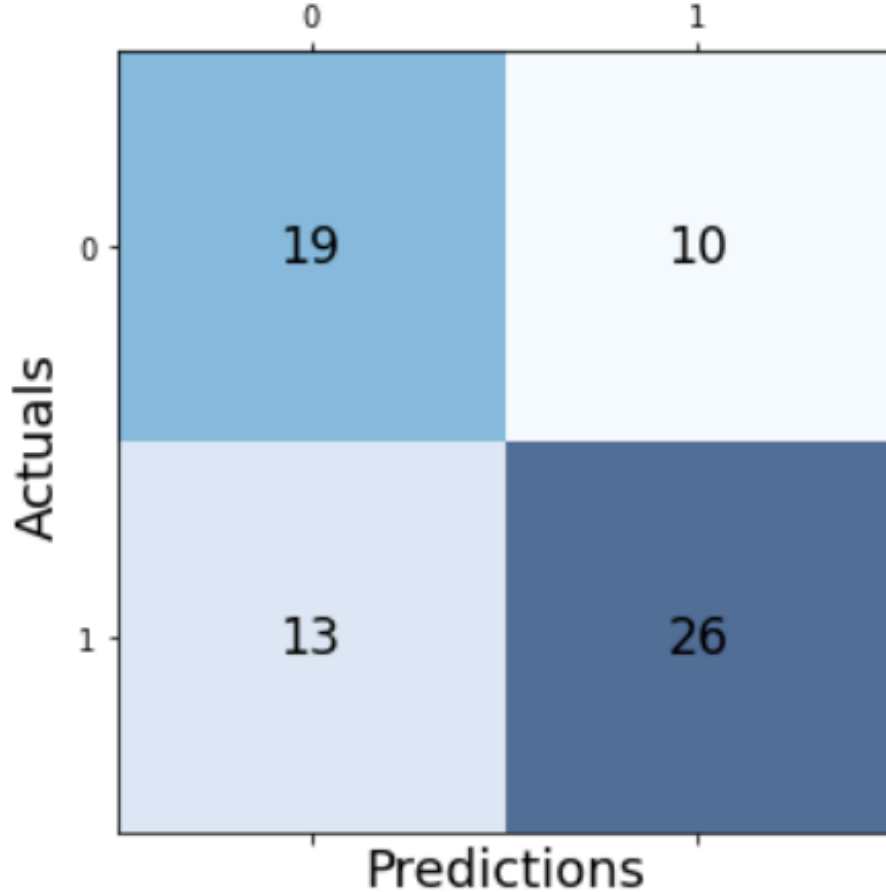


k-NN (Under Sampling)

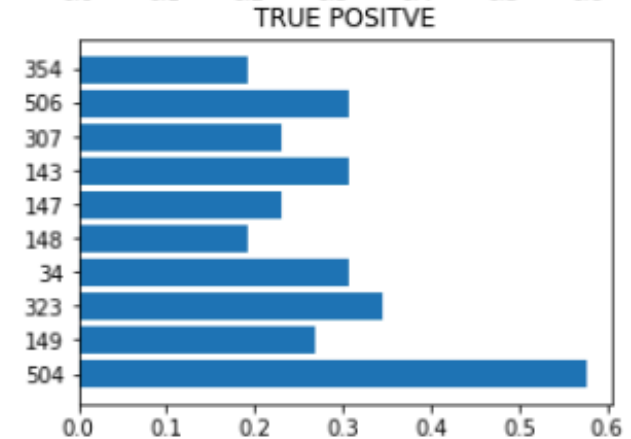
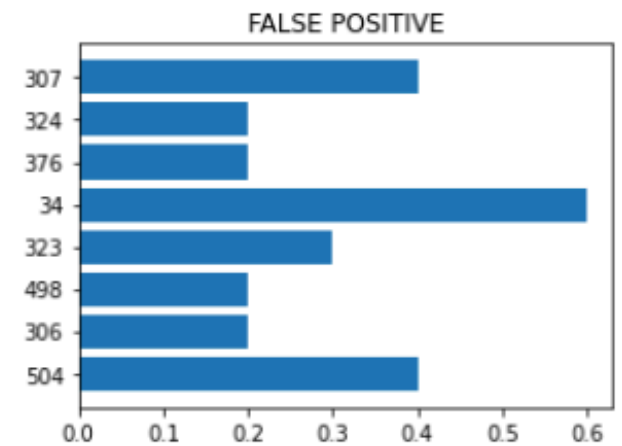
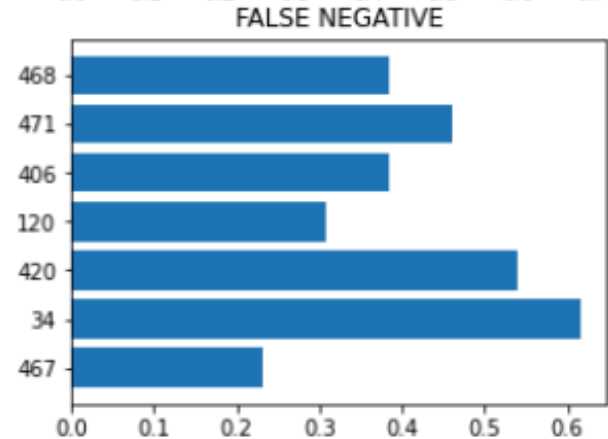
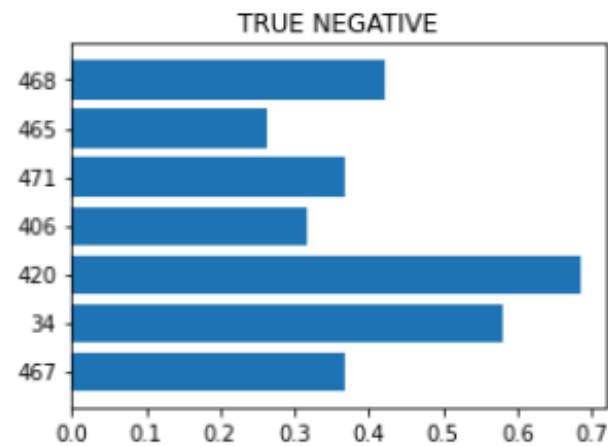


# MODELS USING UNDERSAMPLED DATA

Confusion Matrix



Bernoulli Naive Bayes(Under Sampling)



# MODEL COMPARISON

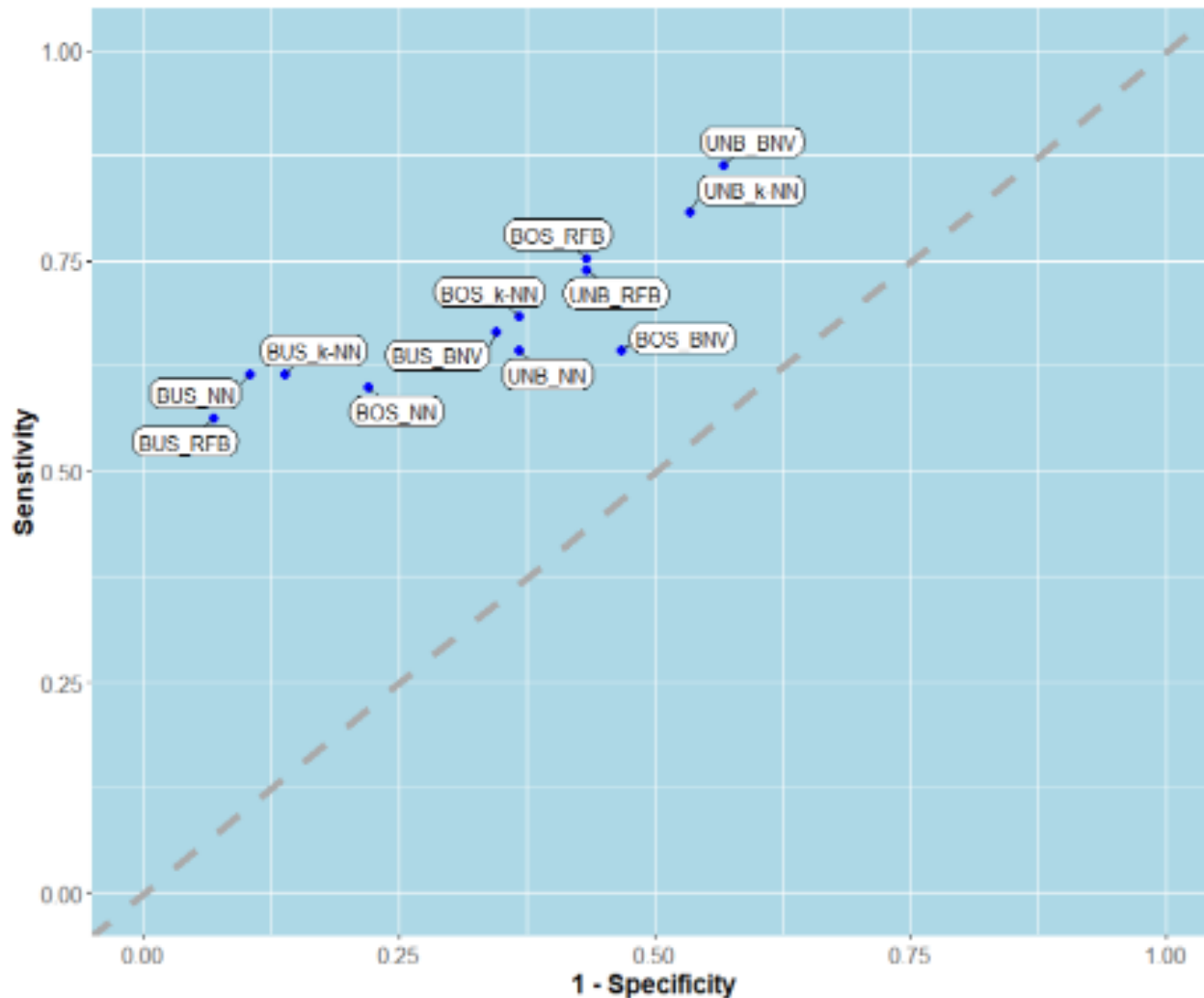


Fig: ROC plot for comparing models

UNB = Unbalanced; BUS = Balanced Under Sampling;  
BOS = Balanced Over Sampling)

- From the ROC curve of all the models (trained with IPC sections only), it is evident that there is a trade-off between specificity and sensitivity,
- For the unbalanced data Sensitivity is high but Specificity is low.
- For balanced data, in case of under sampling, we achieved high Specificity but Sensitivity is low.
- For the models with Over Sampling, both the sensitivity and specificity lie in between.



# EVALUATION METRICS FOR ALL MODELS (IPCS)

| Data Types     | MODEL                 | Sensitivity | Specificity | Accuracy | Precision | F-Score |
|----------------|-----------------------|-------------|-------------|----------|-----------|---------|
| Unbalanced     | Bernoulli Naive Bayes | 0.863       | 0.4333      | 0.7268   | 0.7875    | 0.8235  |
|                | K-NN                  | 0.8082      | 0.4666      | 0.7099   | 0.7875    | 0.7972  |
|                | Neural Network        | 0.6438      | 0.6333      | 0.6367   | 0.8125    | 0.7178  |
|                | Bagging               | 0.7397      | 0.5666      | 0.6799   | 0.8059    | 0.7714  |
| Over Sampling  | Bernoulli Naive Bayes | 0.6438      | 0.5333      | 0.6166   | 0.7704    | 0.7014  |
|                | k-NN                  | 0.6849      | 0.6333      | 0.6859   | 0.8196    | 0.7462  |
|                | Neural Network        | 0.6         | 0.7808      | 0.7211   | 0.826     | 0.8028  |
|                | Bagging               | 0.7534      | 0.5666      | 0.6789   | 0.8088    | 0.7801  |
| Under Sampling | Bernoulli Naive Bayes | 0.6666      | 0.6551      | 0.6595   | 0.7222    | 0.6933  |
|                | k-NN                  | 0.5641      | 0.9655      | 0.7369   | 0.9565    | 0.7096  |
|                | Neural Network        | 0.6153      | 0.8965      | 0.7293   | 0.8889    | 0.7272  |
|                | Bagging               | 0.5641      | 0.931       | 0.7214   | 0.9167    | 0.6984  |

Table : Confusion Matrix Metrics for all the models

# COMPARISON OF MODELS BASED ON IPCS

Table: common IPCs for all models for each type of datasets

|                                      | Dataset        | Correctly classified as Bail | Correctly classifies as No Bail | Misclassified as Bail       | Misclassified as No Bail |
|--------------------------------------|----------------|------------------------------|---------------------------------|-----------------------------|--------------------------|
| B<br>A<br>L<br>A<br>N<br>C<br>E<br>D | Unbalanced     | 506,504,323,307,143,147      | 420,468,467                     | 506,504,307,147             | 420,406,467              |
|                                      | Over Sampling  | 506,504,323,307,143,147,149  | 420,468,467,471,406,201         | 506,504,307,147,323,148,149 | 420,468,471,504,307,506  |
|                                      | Under Sampling | 506,504,323,307,143,147,149  | 420,468,467,471,406             | NA                          | 420,468,471,406,120      |

| U n b a l a n c e d |                         |
|---------------------|-------------------------|
| 420,467,468         | 506,504,307,147         |
| 420,467,406         | 506,504,307,323,143,147 |

| O v e r   S a m p l i n g |                             |
|---------------------------|-----------------------------|
| 420,468,471,467,406,201   | 506,504,307,323,147,148,149 |
| 420,468,471,506,504,307   | 506,504,307,323,147,149,143 |

| U n d e r   S a m p l i n g |                             |
|-----------------------------|-----------------------------|
| 420,468,471,406,467         | NA                          |
| 420,468,471,406,120         | 506,504,323,307,143,147,149 |

Table: common IPCs between misclassified and correctly classified cases

# CONCLUSION

- The interpretation of the result (for models trained using IPC sections) are made assuming the IPCs are independent. So considering that, we are able find out pattern from the data that few IPCs are always coming into picture for discriminating the Bail and No Bail files.
- However we can also see that many times bail files are getting misclassified as No Bail, which might be because of, in practical scenario, many cases with serious charges like 420 (No Bailable) 376 (rape), which would have been generally in No bail files, are also getting Bail.
- This practically make sense as even if the charges are serious, but if the accuse have enough evidence they get bail easily, but as our models are solely dependent on IPCs, it is decreasing the power of our model in discrimination.
- There are a lot of scopes in further exploration like,
  - a. including text from police charge sheet also in study to check whether that increases the discrimination,
  - b. Finding out if there is any joint behaviour of IPCs, as in case of over sampled dataset, 506, 504 and 307 is coming in False Bail files, as these are prominently present True Bail, but it is also going to False No Bail files.

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**Thank You**